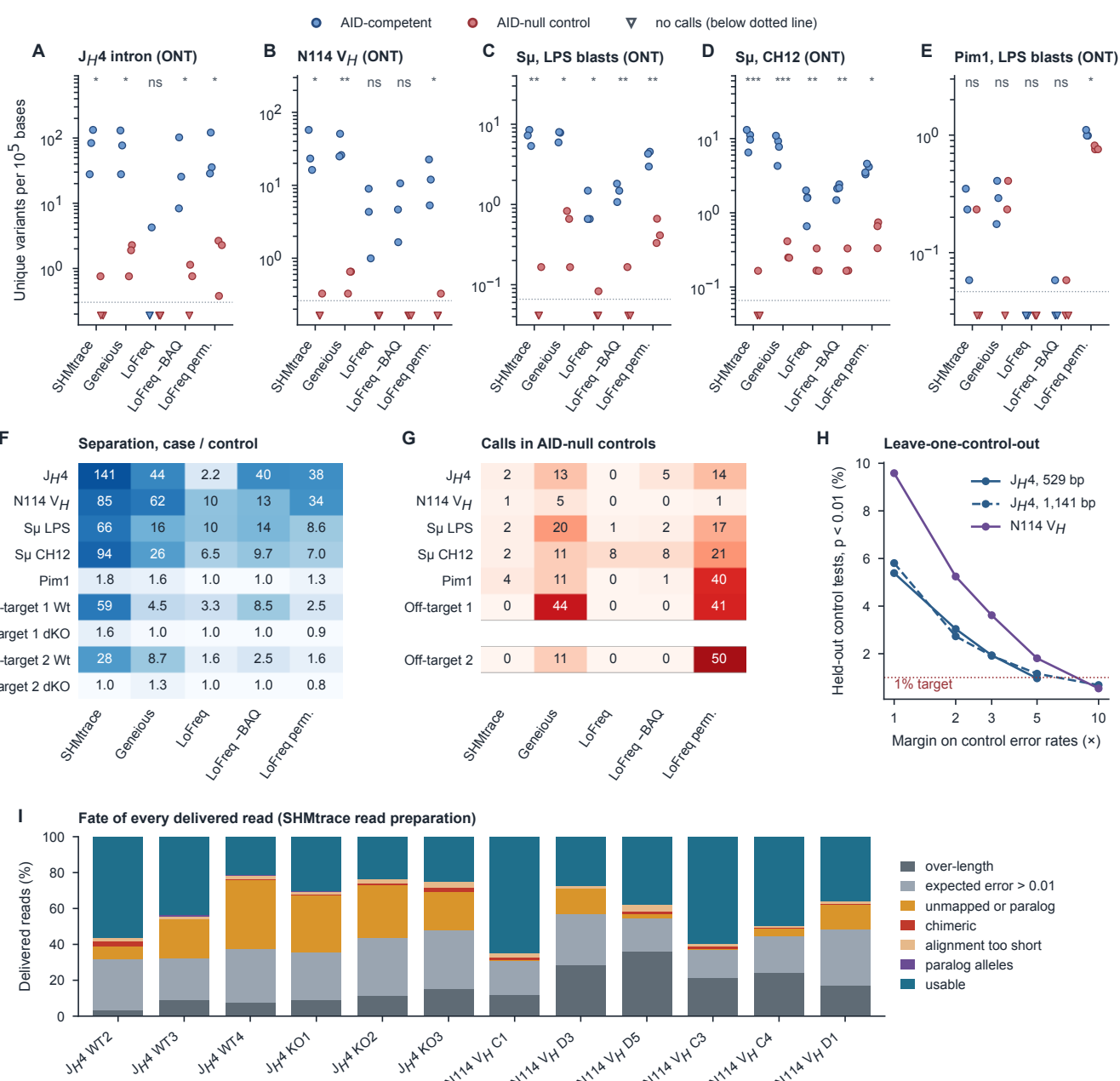

AID-competent
AID-null control
no calls (below dotted line)
A
J_H4 intron (ONT)
B
N114 V_H (ONT)
C
Sμ, LPS blasts (ONT)
D
Sμ, CH12 (ONT)
E
Pim1, LPS blasts (ONT)
Unique variants per 10^5 bases
SHMtrace
Geneious
LoFreq
LoFreq −BAQ
LoFreq perm.
F
Separation, case / control
J_H4 141 44 2.2 40 38
N114 V_H 85 62 10 13 34
Sμ LPS 66 16 10 14 8.6
Sμ CH12 94 26 6.5 9.7 7.0
Pim1 1.8 1.6 1.0 1.0 1.3
-target 1 Wt 59 4.5 3.3 8.5 2.5
arget 1 dKO 1.6 1.0 1.0 1.0 0.9
-target 2 Wt 28 8.7 1.6 2.5 1.6
arget 2 dKO 1.0 1.3 1.0 1.0 0.8
G
Calls in AID-null controls
J_H4 2 13 0 5 14
N114 V_H 1 5 0 0 1
Sμ LPS 2 20 1 2 17
Sμ CH12 2 11 8 8 21
Pim1 4 11 0 1 40
Off-target 1 0 44 0 0 41
Off-target 2 0 11 0 0 50
H
Leave-one-control-out
Held-out control tests, p < 0.01 (%)
J_H4, 529 bp
J_H4, 1,141 bp
N114 V_H
1% target
Margin on control error rates (×)
I
Fate of every delivered read (SHMtrace read preparation)
Delivered reads (%)
over-length
expected error > 0.01
unmapped or paralog
chimeric
alignment too short
paralog alleles
usable
J_H4 WT2
J_H4 WT3
J_H4 WT4
J_H4 KO1
J_H4 KO2
J_H4 KO3
N114 V_H C1
N114 V_H D3
N114 V_H D5
N114 V_H C3
N114 V_H C4
N114 V_H D1